

PROJECT BUTTON HR–V0 ELECTRICAL V3 WATCHDOG–INTERLOCK CANDIDATE

P1.19 layout preserved; two ordinary watchdog contacts interrupt separate SRA1 input returns; zero safety credit.

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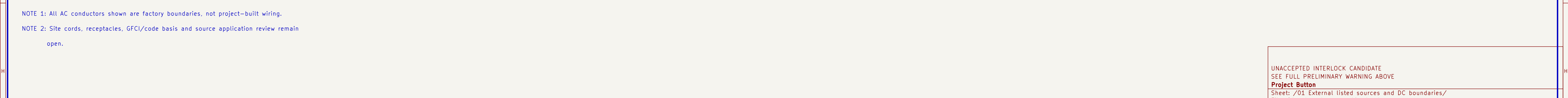
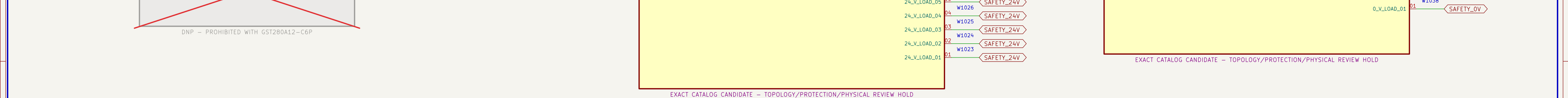
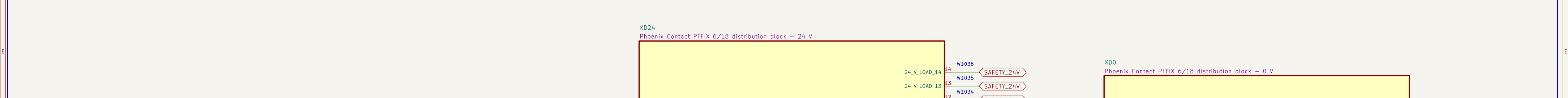
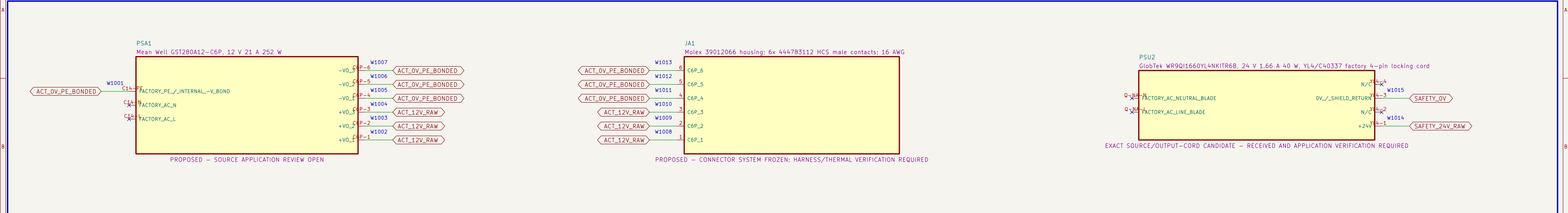
File: 10_actuator_interfaces.kicad_sch

11 Watchdog PCB external connectors

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12 Watchdog PCB test access

File: 12_watchdog_pcb_test_access.kicad_sch



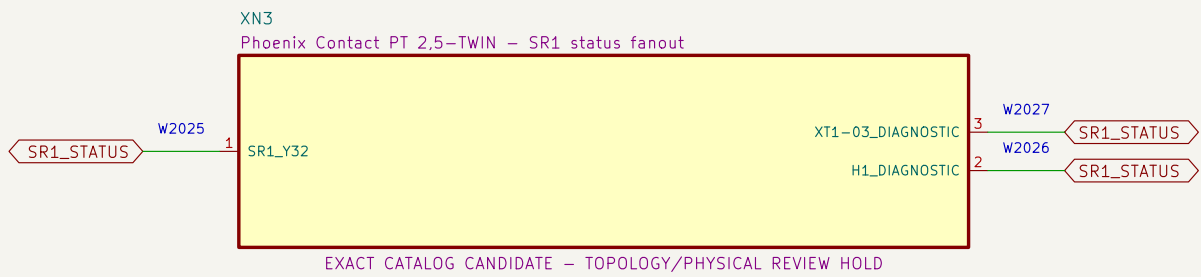
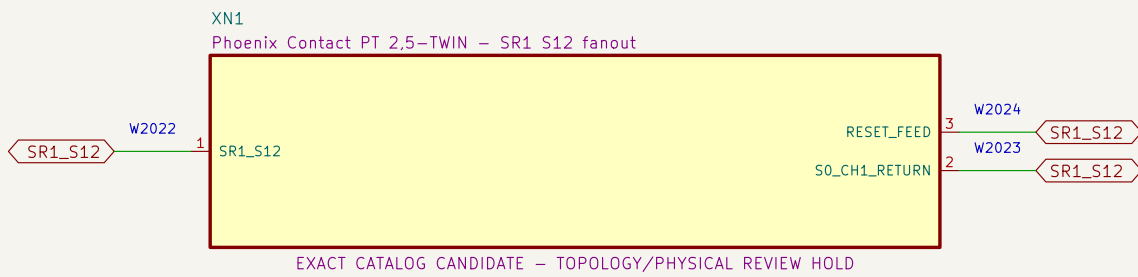
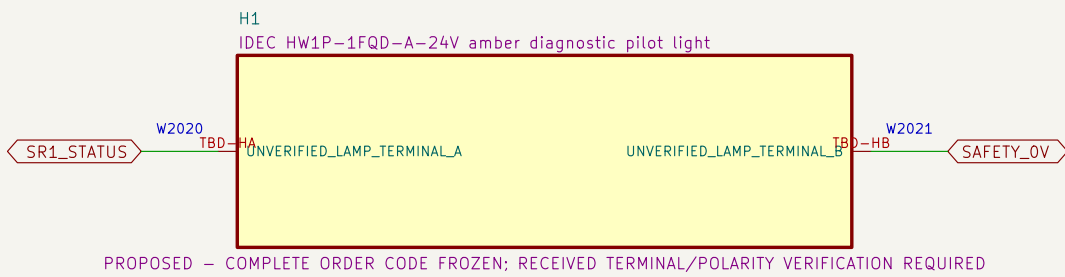
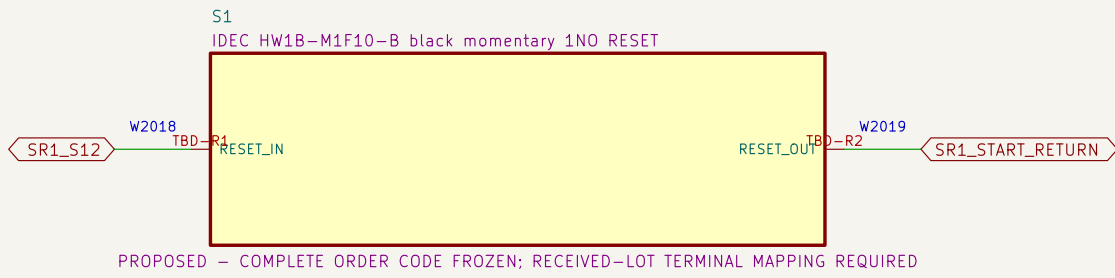
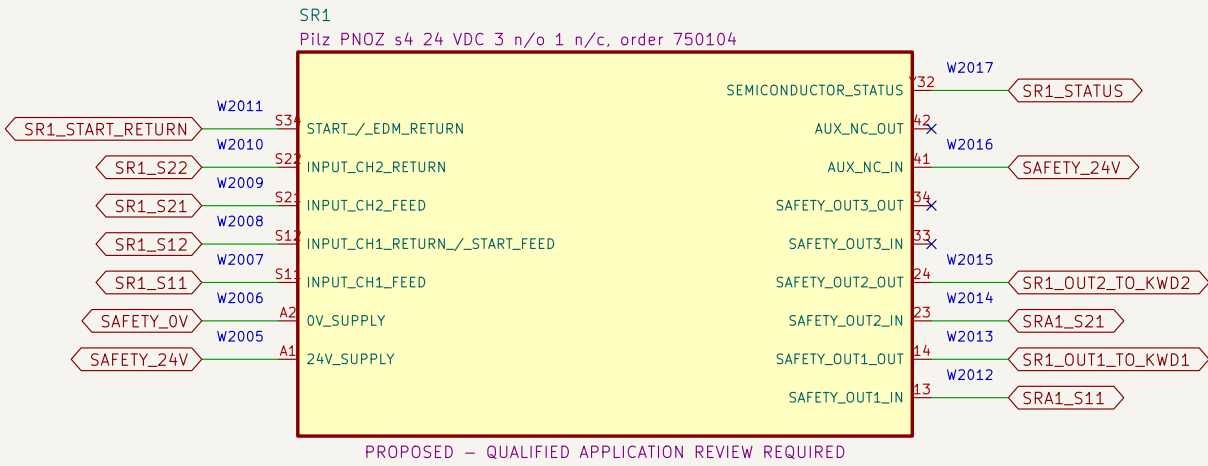
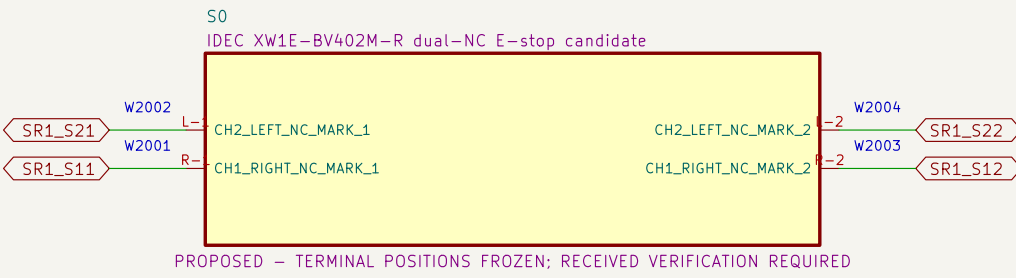
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Title: PB HR-V0 ELEC P1.20 - 01		
Size: A2	Date: 2026-08-11	Rev: P1.20

Size: A2	Date: 2026-08-11	Rev: P1.20
KiCad E.D.A. 10.0.5		Id: 2/13

10	11
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02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains only one S0 NC contact; RESET cannot energize K1/K2.

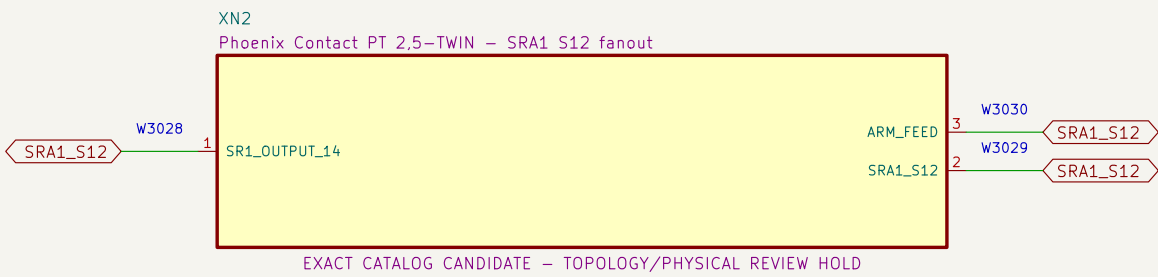
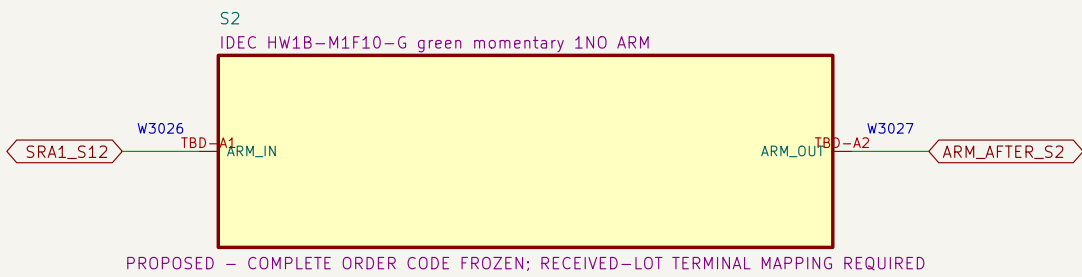
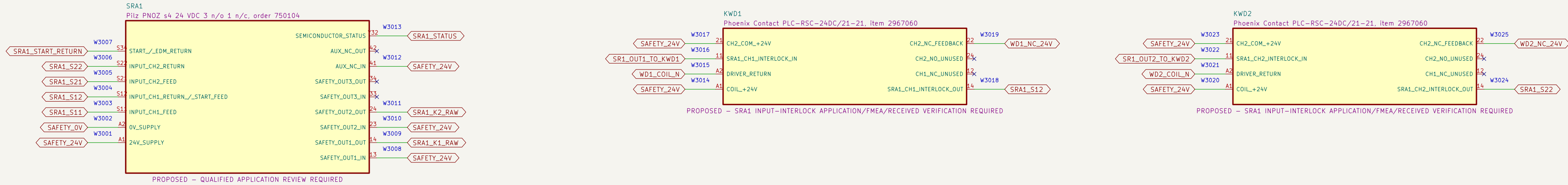


NOTE 1: S0 directly completes both SR1 input returns; ordinary KWD terminals are absent from both loops.

NOTE 2: Heartbeat loss opens both SRA1 input returns while SR1 remains powered; recovery alone cannot restore the monitored ARM stage.

03 Distinct ARM and dual-channel watchdog interlock

Separate ordinary KWD contacts interrupt the two SRA1 input returns; both stages retain zero safety credit.

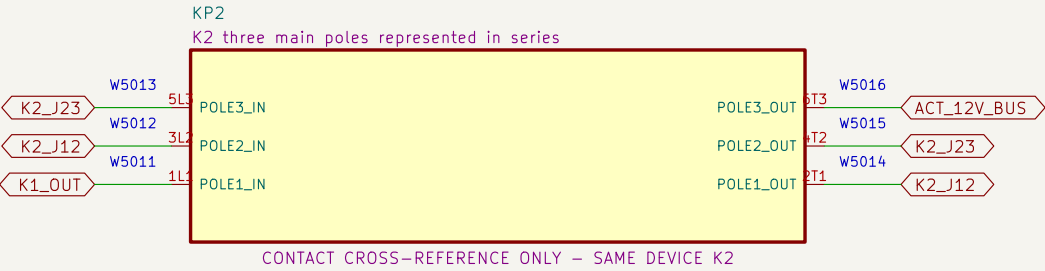
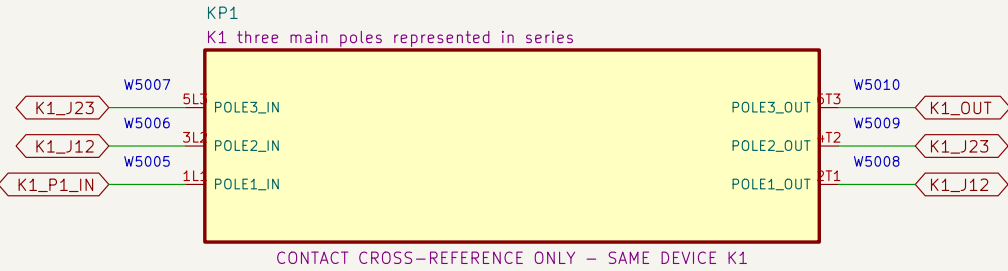
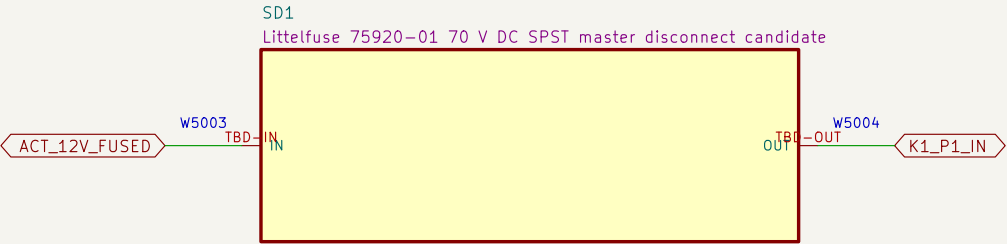
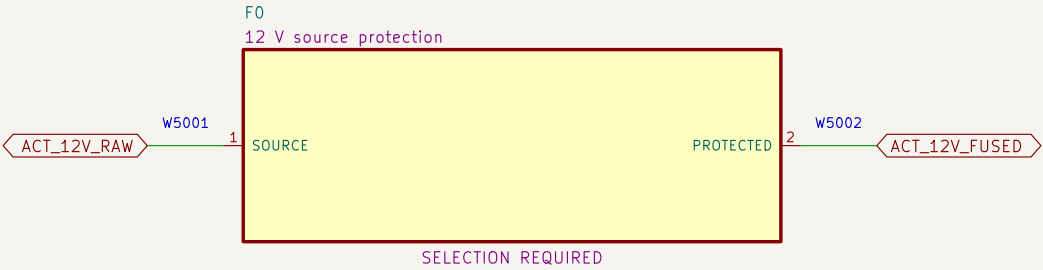


NOTE 1: After a successful watchdog dropout: cause healthy -> both KWD contacts restored
-> SR1 remains SAFE_READY -> distinct monitored ARM press/release -> K1/K2 may energize.

NOTE 2: KWD1 interrupts SR1:14 to SRA1:S12 and KWD2 interrupts SR1:24 to SRA1:S22. One welded contact is defeated by the other opening: dual/common-cause failures remain hazardous and receive zero safety credit.

05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

UNACCEPTED INTERLOCK CANDIDATE
SEE FULL PRELIMINARY WARNING ABOVE

Project Button

Sheet: /05 Redundant actuator–power interruption/
File: 05_actuator_interruption.kicad_sch

Title: PB HR–V0 ELEC P1.20 – 05

Size: A3

Date: 2026–08–11

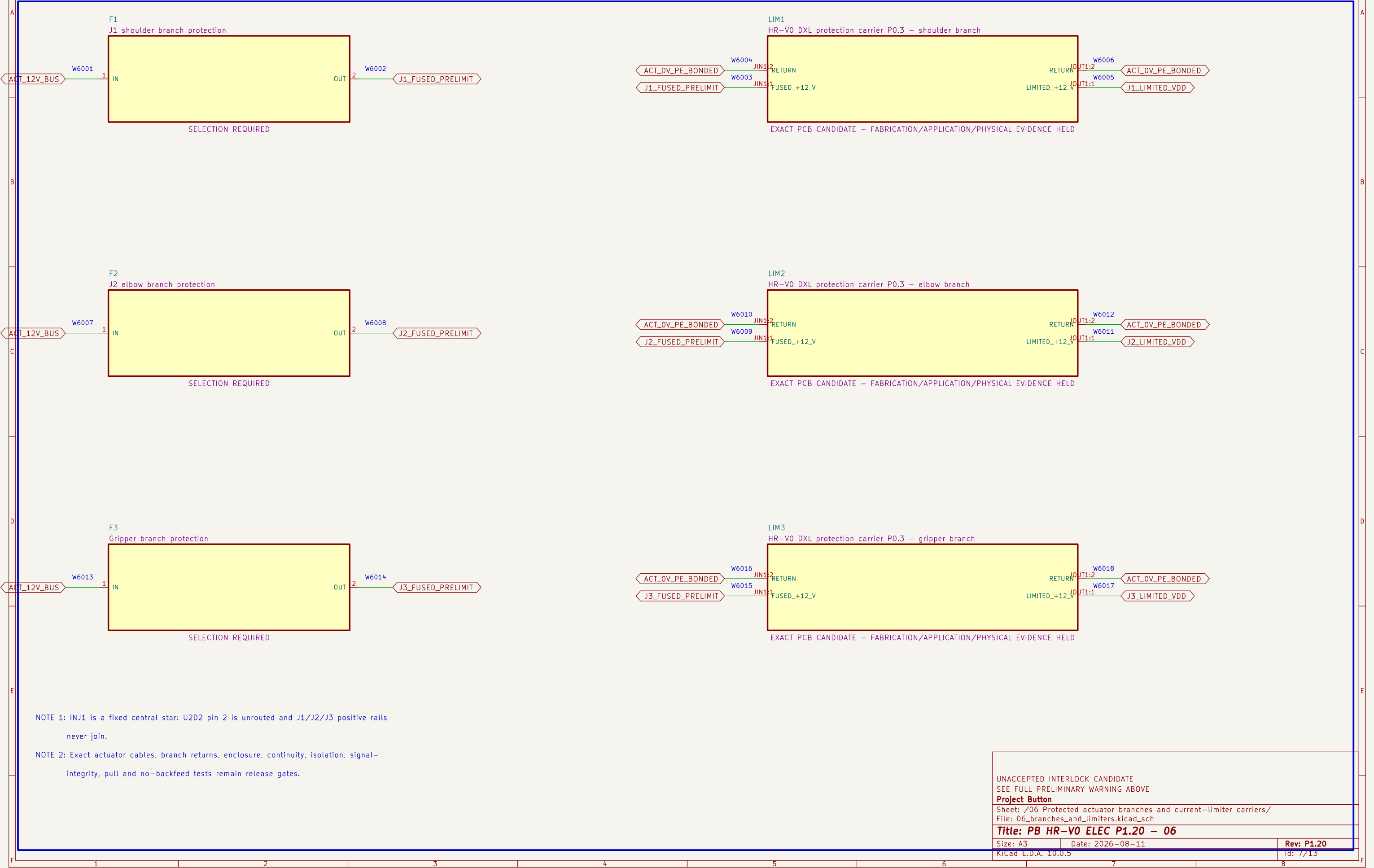
Rev: P1.20

KiCad E.D.A. 10.0.5

Id: 6/13

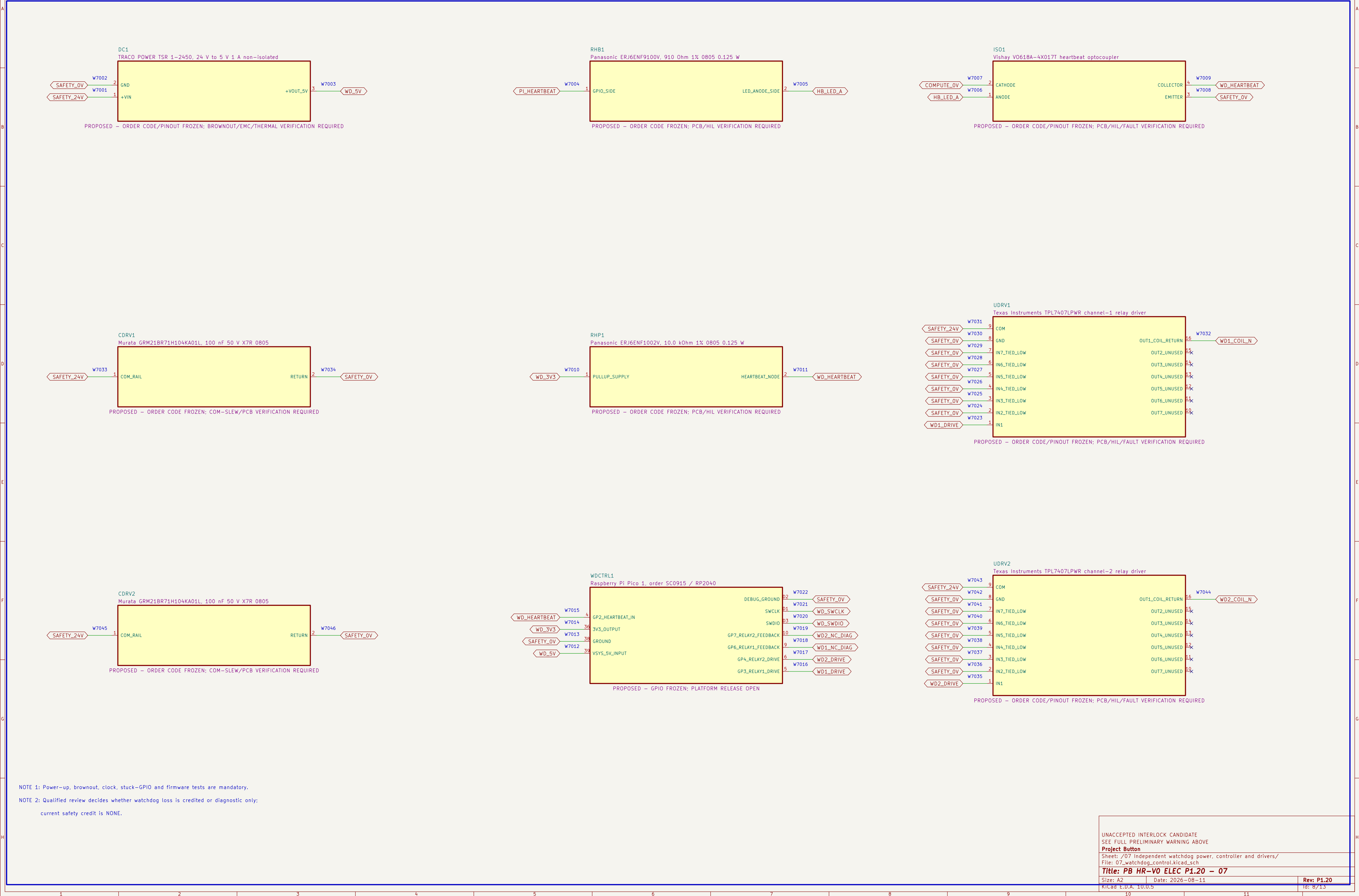
06 Protected actuator branches and current-limiter carriers

Each fuse output and carrier output has a distinct positive rail; physical protection performance remains unproved.



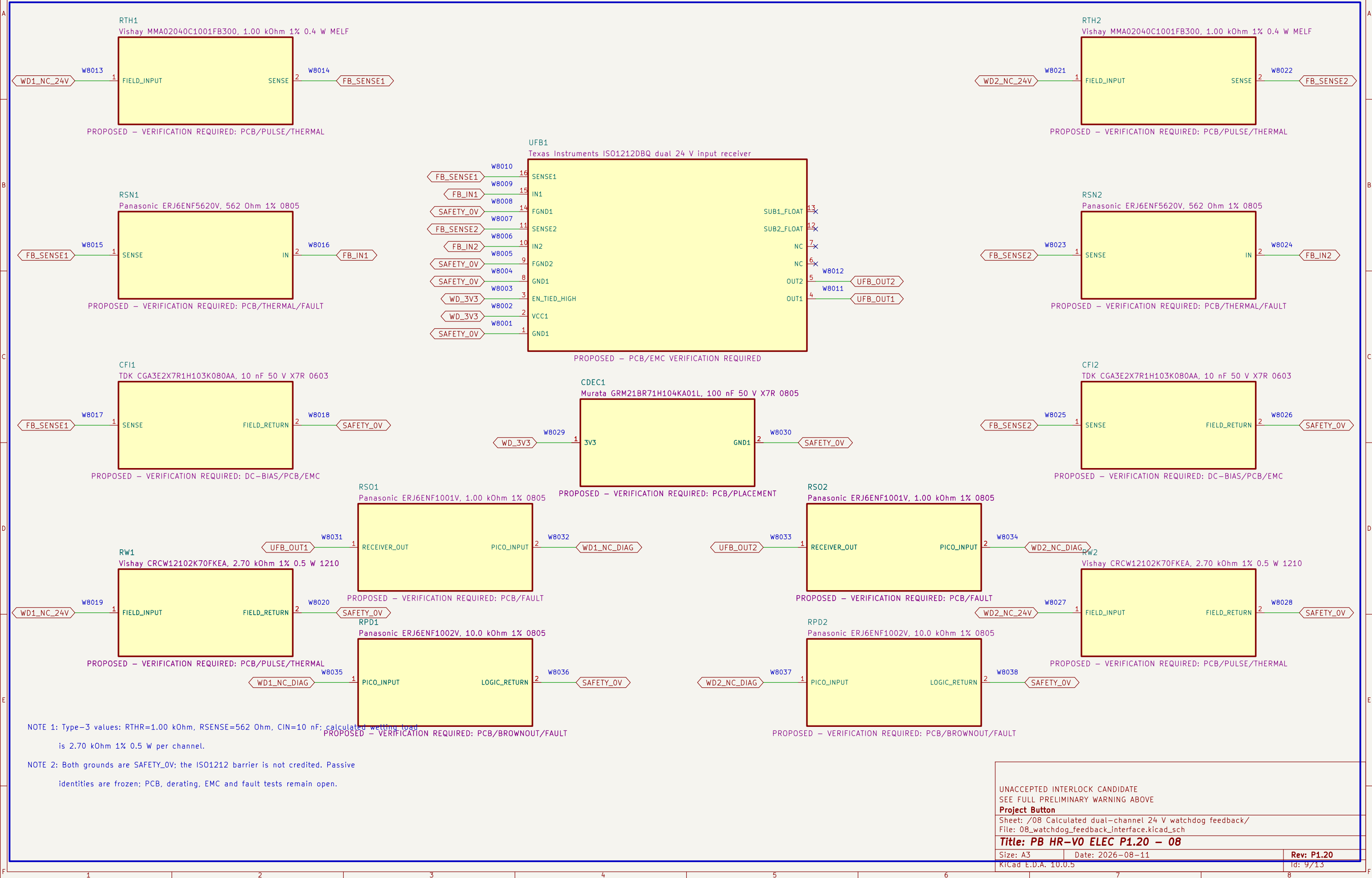
07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



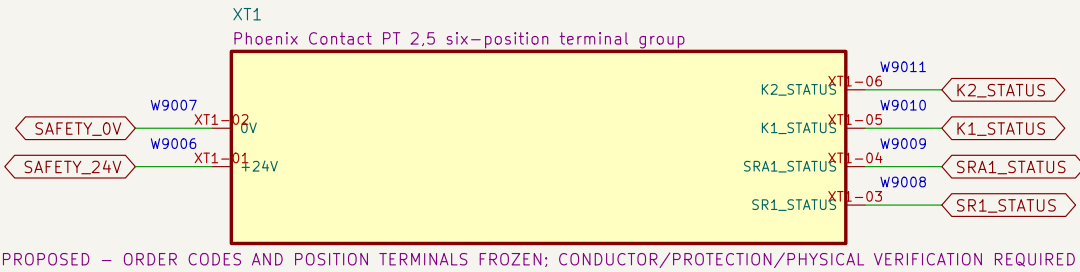
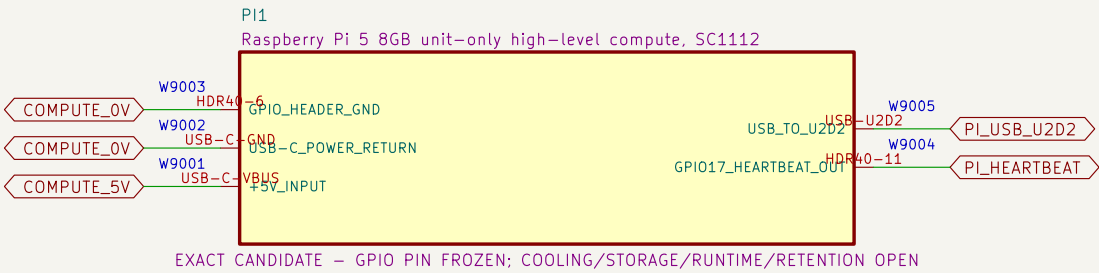
08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



09 Compute and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.

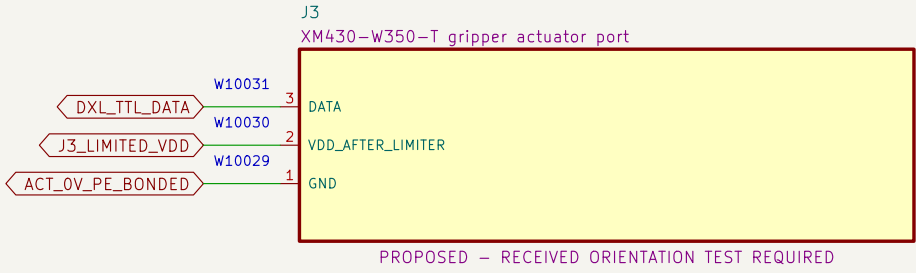
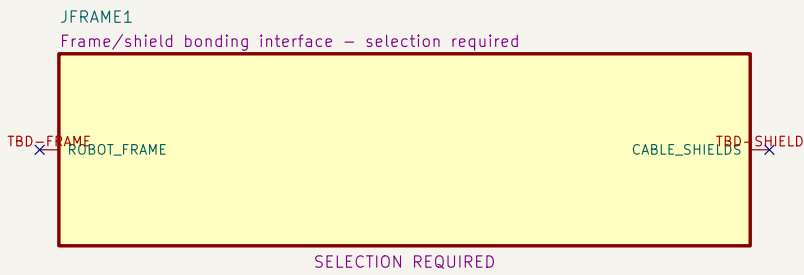
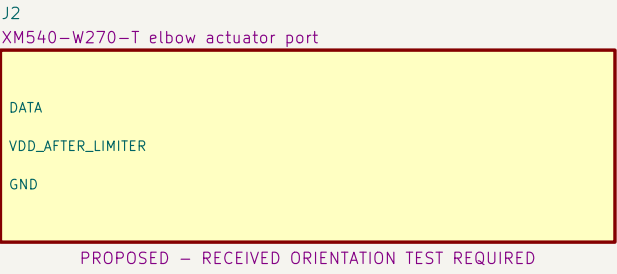
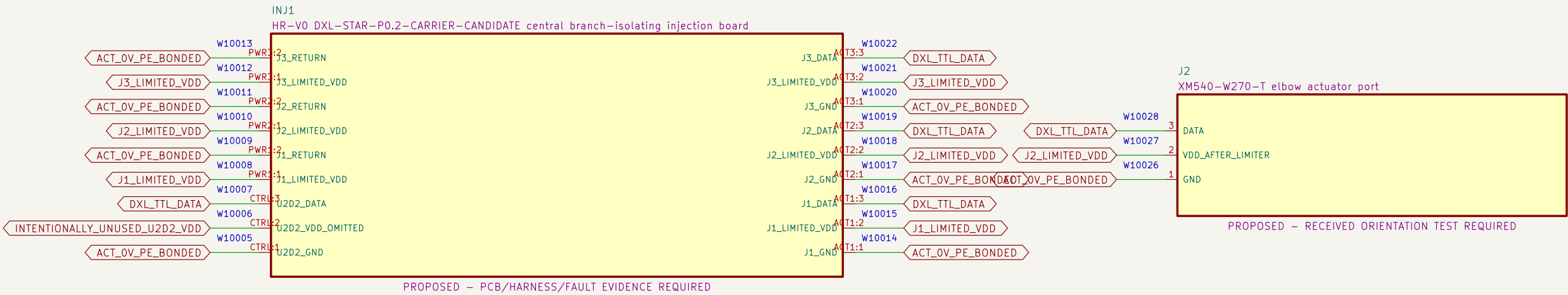
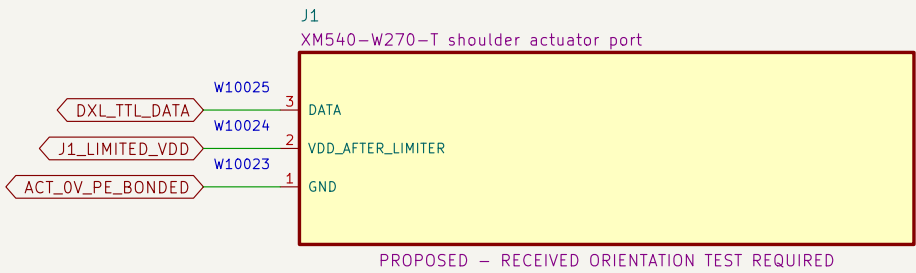
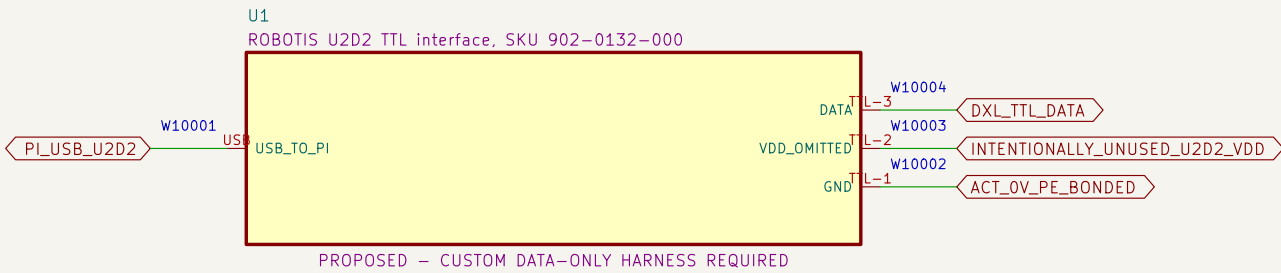


- NOTE 1: No installed debug connector exists; watchdog programming uses only TP15/TP16/TP2 under a future controlled unpowered-fixtured procedure.
- NOTE 2: Debug connection, halt, flash and disconnect must not enable outputs, back-power the board or bypass a protective function.
- NOTE 3: Heartbeat cable, GPIO runtime, timing, terminal family, markers, conductor range, torque and enclosure layout remain selection required.

UNACCEPTED INTERLOCK CANDIDATE SEE FULL PRELIMINARY WARNING ABOVE Project Button		
Sheet: /09 Compute and control terminals/ File: 09_compute_and_control_terminals.kicad_sch		
Title: PB HR-V0 ELEC P1.20 – 09		
Size: A3	Date: 2026-08-11	Rev: P1.20
KiCad E.D.A. 10.0.5		Id: 10/13

10 U2D2, DXL star, actuator ports and bonding boundary

U2D2 pin 2 is omitted; DXL–STAR accepts three distinct limited positive rails and routes DATA/return.

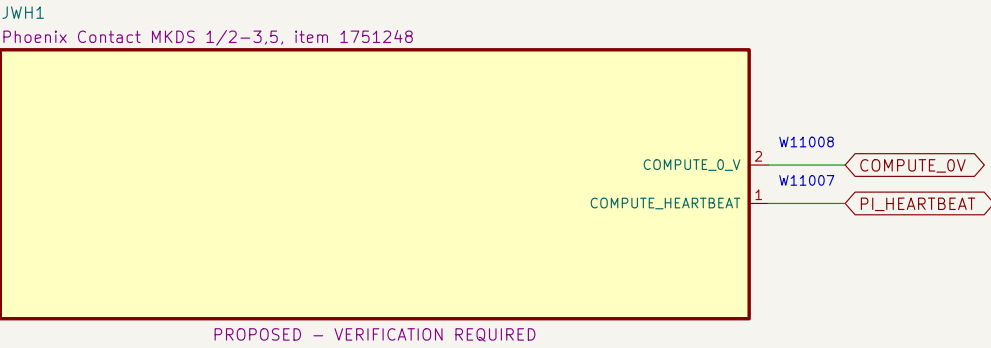
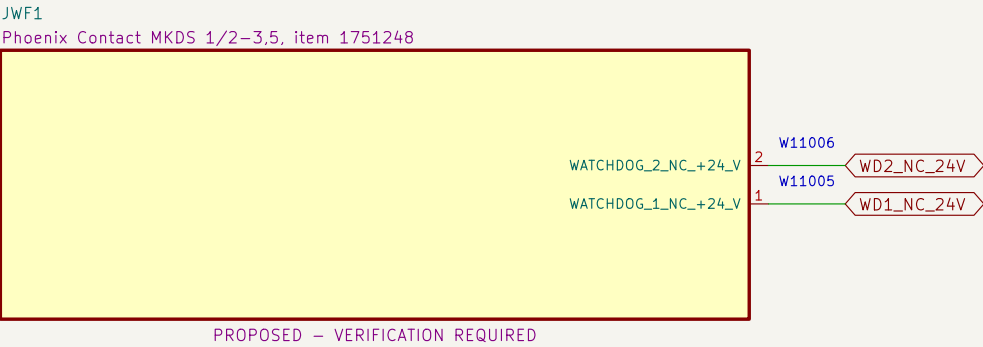
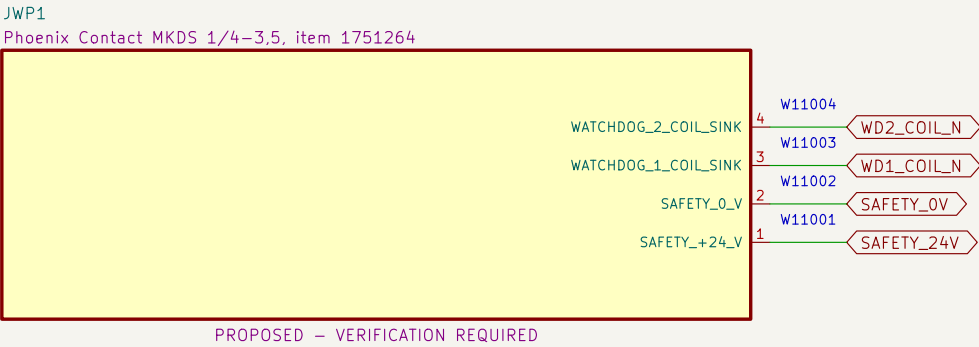


NOTE 1: The V0 bus is TTL; HR–30 RS–485 remains a separate architecture.

NOTE 2: Do not bond frame or shields to OV/PE until EMC and parallel–path review accepts the exact implementation.

11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.



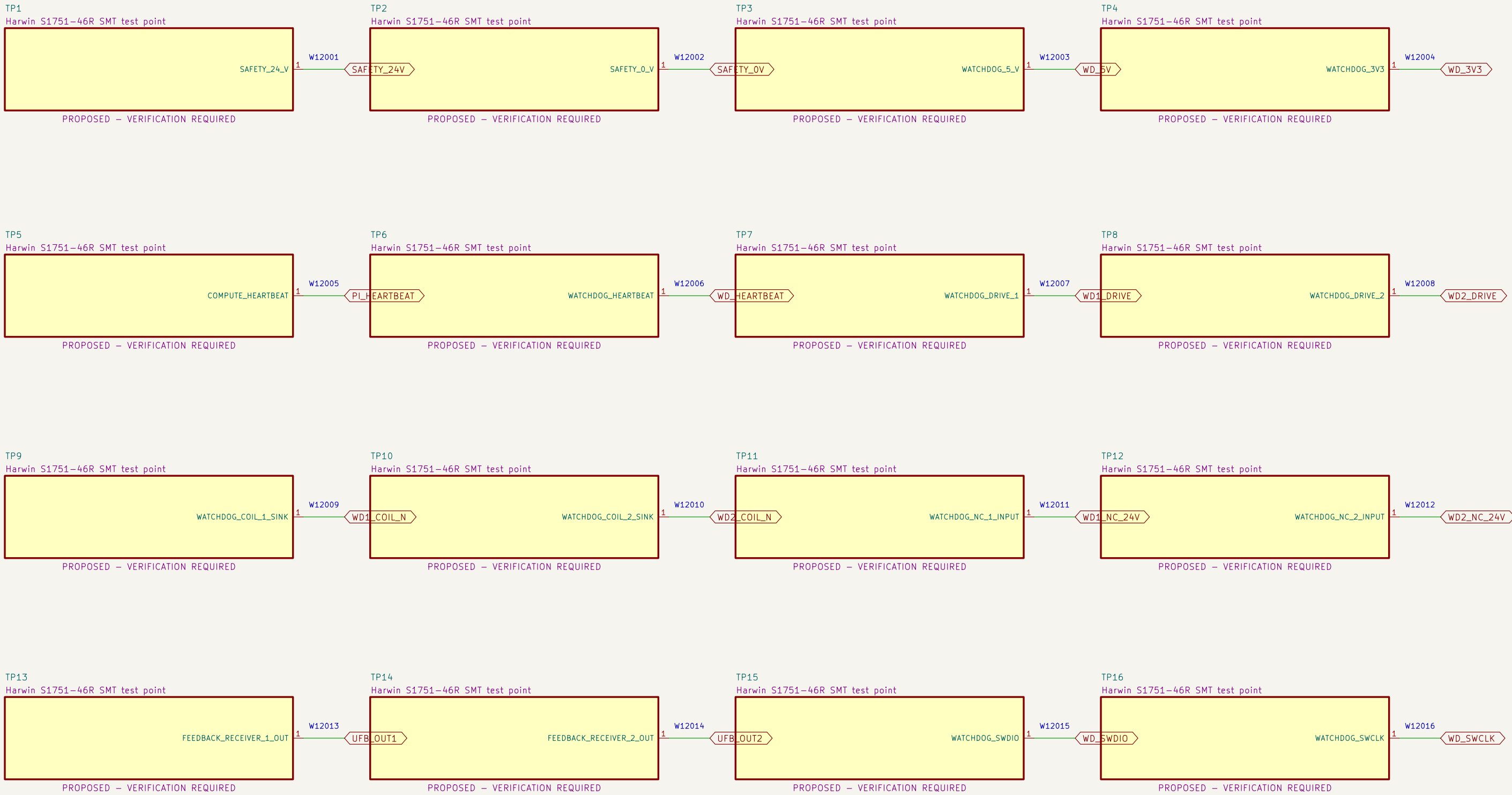
NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

UNACCEPTED INTERLOCK CANDIDATE SEE FULL PRELIMINARY WARNING ABOVE		
Project Button		
Sheet: /11 Watchdog PCB external connectors/ File: 11_watchdog_pcb_connectors.kicad_sch		
Title: PB HR–VO ELEC P1.20 – 11		
Size: A3	Date: 2026–08–11	Rev: P1.20
KiCad E.D.A. 10.0.5	Id: 12/13	

12 Watchdog PCB test access

Exact clip-compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.